

Lecture 7 - Part V - Extras 2

- Expenditure minimization
- Price Indexes

ECON 101

Fall 2020



Consumer Problem.

Given preferences, prices and income,

Given: $U(x_1, x_2)$, p_1, p_2 , I

Consumer Problem: maximize $U(x_1, x_2)$

↳ To choose the best (x_1, x_2)

bundle, according to consumer preferences, in my budget-set.

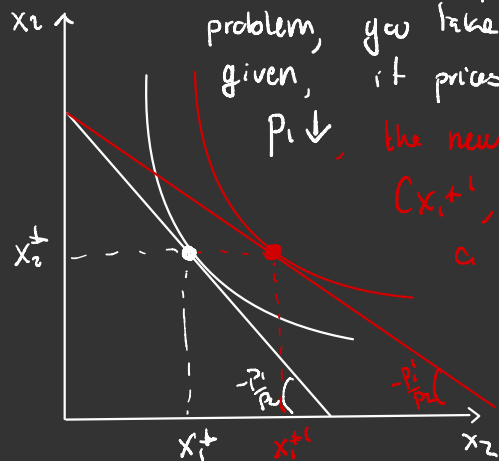
s.t. $\underbrace{p_1 x_1 + p_2 x_2}_{\text{Expenditure}} \leq I$

At (x_1^*, x_2^*) : $MRS = \frac{p_1}{p_2}$, in this

problem, you take p_1, p_2, I as given, if prices change, say

$p_1 \downarrow$, the new bundle

$(x_1^{*'}, x_2^{*'})$ achieves a higher indifference curve.



Consumer Problem (Expenditure Minimization)

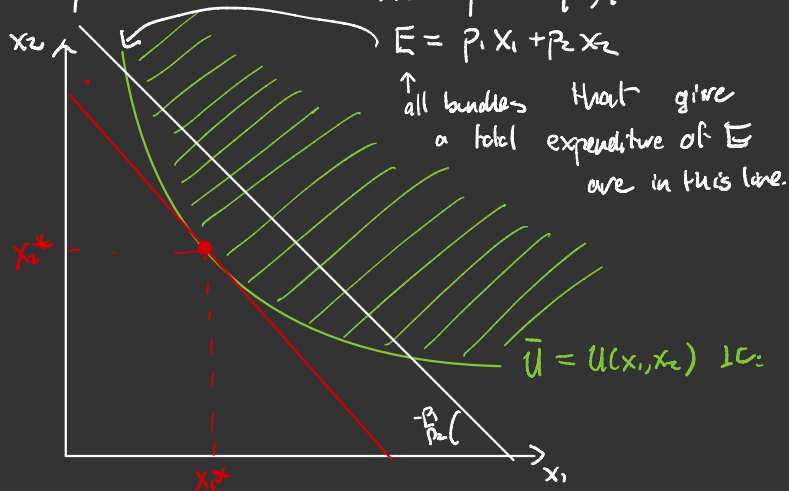
Given preferences, prices and a fixed level of utility:

$U(x_1, x_2)$, p_1, p_2 , $\bar{U}$ (level of utility $\bar{U}$)

Expenditure minimization: minimize $p_1 x_1 + p_2 x_2$
s.t. $U(x_1, x_2) \geq \bar{U}$

↓
To choose the cheapest bundle that achieves the required utility level $\bar{U}$

Given prices p_1, p_2



Expenditure Problem of the consumer: parameters: $p_1, p_2, \bar{u}$

$$\begin{aligned} &\text{minimize}_{x_1, x_2} \quad p_1 x_1 + p_2 x_2 \\ &\text{s.t.} \quad U(x_1, x_2) \geq \bar{u} \end{aligned}$$

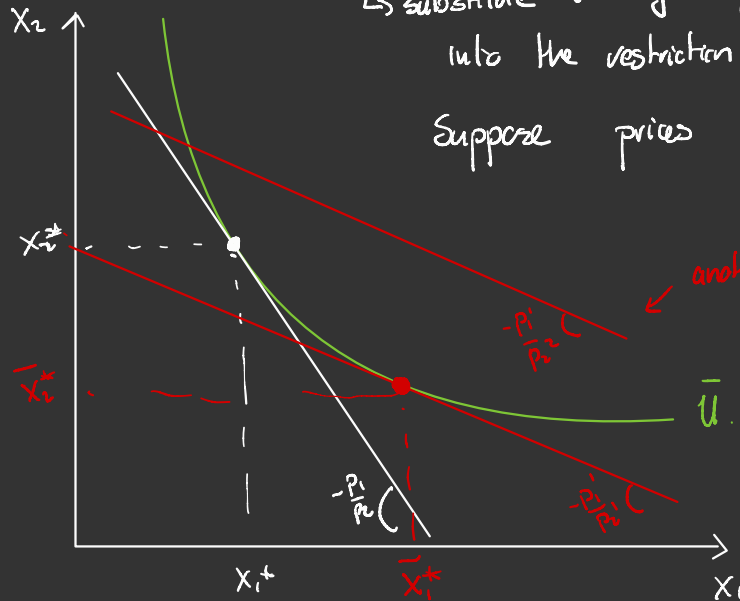
Solutions to this problem, when prices change will give an optimal bundle that achieves exactly the original utility, but at different prices.

- Optimality condition: $MRS = \frac{p_1}{p_2}$

↳ substitute what you get

into the restriction $U(x_1, x_2) = \bar{u}$

Suppose prices change: (p'_1, p'_2)



another expenditure line: $E = p'_1 x_1 + p'_2 x_2$

(x_1^*, x_2^*)
 (x_1', x_2') } both achieve utility $\bar{u}$

	2000	2010
Price of books	\$20/book	\$107.5/book
Number of books	15	9
Price of food	\$2.00/lb.	\$3.00/lb.
Pounds of food	100	215
Expenditure	\$500	\$1612.50

Suppose the consumer has utility function

$$U(F, B) = F^{0.4} B^{0.6} \quad (MRS = \frac{0.4 B}{0.6 F} = \frac{2}{3} \frac{B}{F})$$

In 2000, assume the consumer had 500 in income,
 $(P_F, P_B) = (2.00, 20)$,

$$\max_{(F, B)} F^{0.4} B^{0.6}$$

$$\text{s.t.} \quad 2F + 20B = 500$$

$$\text{Optimal bundle satisfies:} \quad \frac{2}{3} \frac{B}{F} = \frac{2}{20} = \frac{1}{10} \Rightarrow B = \frac{3}{2} \cdot \frac{1}{10} \cdot F$$

$$B = \frac{3}{20} \cdot F$$

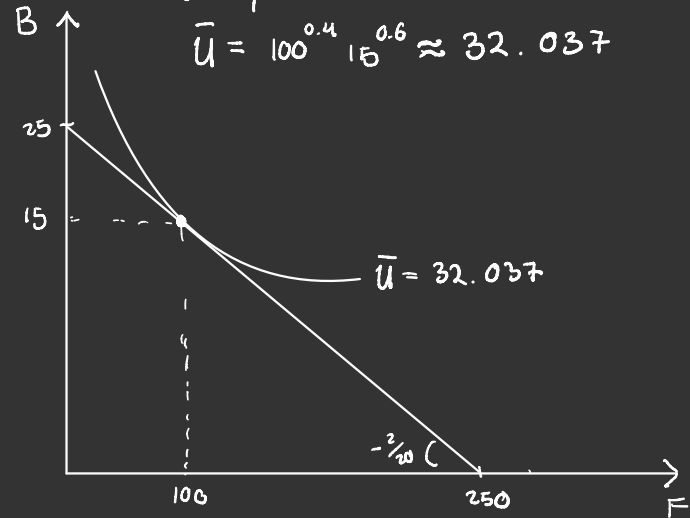
$$\text{Plug the previous into B.L.:} \quad 2F + 20\left(\frac{3}{20}\right)F = 500$$

$$\Rightarrow 2F + 3F = 5F = 500 \Rightarrow F = 100, \quad B = \frac{3}{20} \cdot 100 = 15$$

Let $A = (100, 15)$, in the year 2000, the consumer chose bundle A when facing prices $(P_F, P_B) = (2, 20)$ and having an income of 500.

Utility of bundle A:

$$\bar{U} = 100^{0.4} 15^{0.6} \approx 32.037$$



How much would it take for the consumer at 2010 prices $(P_F, P_B) = (3, 107.5)$ to achieve the same utility $\bar{U}$.

	2000	2010
Price of books	\$20/book	\$107.5/book
Number of books	15	9
Price of food	\$2.00/lb.	\$3.00/lb.
Pounds of food	100	215
Expenditure	\$500	\$1612.50

To fill the table in 2010, we solve for the expenditure minimization problem of the consumer:

$$\begin{aligned} &\text{minimize}_{F, B} \quad 3F + 107.5B \\ &\text{subject to} \quad F^{0.4} B^{0.6} = \bar{U} = 32.037 \end{aligned}$$

$$\text{Optimality condition:} \quad \frac{\frac{2}{3} \frac{B}{F}}{\frac{3}{107.5}} = \frac{6}{215}$$

$$B = \frac{6}{215} \cdot \frac{3}{2} F = \frac{9}{215} F$$

$$B = \frac{9}{215} F$$

$$\text{Plug in } B = \frac{9}{215} F \text{ into}$$

$$32.037 = F^{0.4} B^{0.6}$$

$$32.037 = F^{0.4} \left(\frac{9}{215} F \right)^{0.6}$$

$$32.037 = F^{0.4} \left(\frac{9}{215} \right)^{0.6} F^{0.6}$$

$$32.037 = F \left(\frac{9}{215} \right)^{0.6}$$

$$F = \frac{32.037}{(9/215)^{0.6}} \approx 215$$

$$B = \frac{9}{215} (215) = 9$$

$$\text{Bundle } B = (215, 9)$$

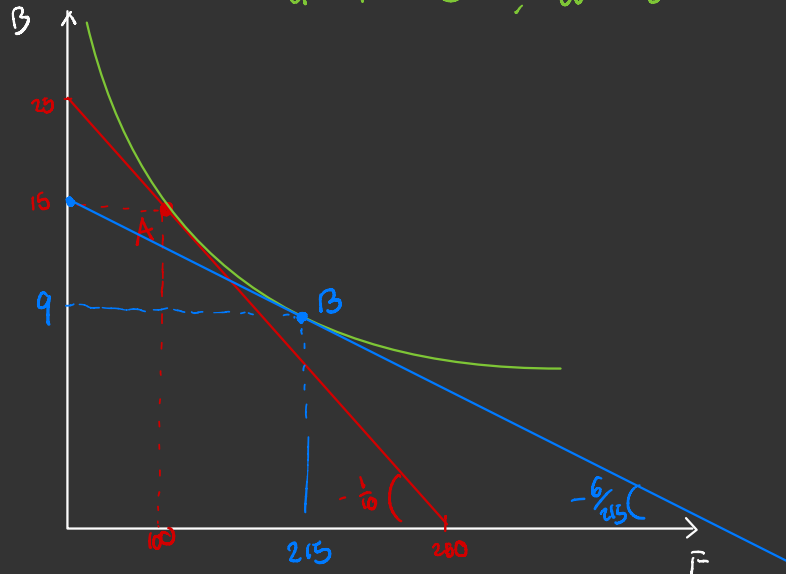
$$U(215, 9) = 215^{0.4} 9^{0.6} \approx 32.037$$

The level of expenditure is calculated as:

$$3(215) + 107.5(9) = 1612.50$$

	2000	2010
Price of books	\$20/book ✓	\$107.5/book ✓
Number of books	15	9
Price of food	\$2.00/lb. ✓	\$3.00/lb. ✓
Pounds of food	100	215
Expenditure	\$500 ✓	\$1612.50 ✓

IC: $\bar{U} = F^{0.4} B^{0.6}$, $\bar{U} \approx 32.037$



Year 2000 $\frac{P_F}{P_B} = \frac{1}{10} = \frac{2}{20}$, $A = (100, 16)$

Year 2010 $\frac{P_F}{P_B} = \frac{3}{107.5} = \frac{6}{215}$, $B = (215, 9)$

Cost of A in 2000 = 500

Cost of B in 2010 = 1612.50

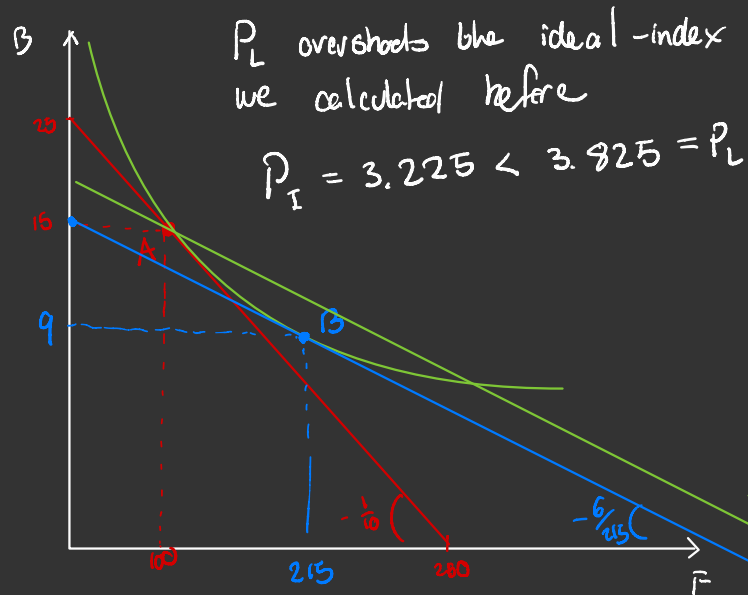
Ideal price-index

$$P_I = \frac{1612.50}{500} = 3.225$$

Ideal because we are computing how much it costs to live in 2010 compared to 2000, while utility level fixed.

Intercept 537.5

	2000	2010
Price of books	\$20/book	\$107.5/book
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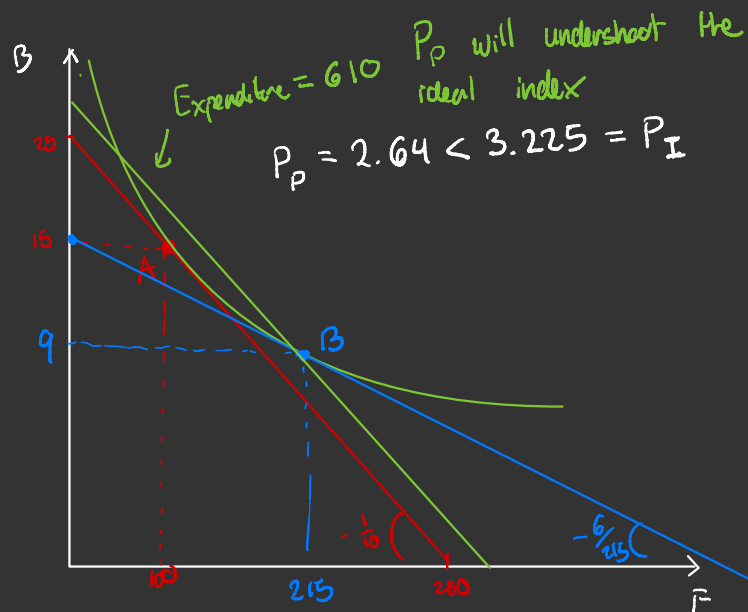
Suppose we consider bundle $A = (100, 15)$ as the base bundle (2000 year the base year).

Cost A at 2010 prices: $3(100) + 107.5(15) = 1,912.5$
 2010: $(P_F, P_B) = (3.00, 107.5)$

$$P_L = \frac{\text{Cost A at 2010 prices}}{\text{Cost A at 2000 prices}} = \frac{1912.5}{500} = 3.825$$

P_L is called a "Laspeyres" price-index,
 fixing a market bundle at a Base year,
 then calculate the ratio of the cost of
 the base basket at current and base year
 prices.

	2000	2010
Price of books	\$20/book	\$107.5/book
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Suppose we instead look at current year (2010) market basket $B = (215, 9)$. year (2000) base year.

Compute the cost of B at current year prices vs base year prices.

Cost of B in 2010 prices: $3(215) + 107.5(9) = 1612.50$

Cost of B in 2000 (base) prices: $2(215) + 20(9) = 610$

$$P_P = \frac{\text{Cost of } B \text{ in 2010}}{\text{Cost of } B \text{ in 2000}} = \frac{1612.50}{610} = 2.64$$

P_P is called the Paasche index. Fix the market bundle at the current year and compute the ratio of its cost at current year prices versus base year prices.

Laspeyres and Paasche Indexes

For a set of goods and services $C = \{c_1, c_2, \dots, c_n\}$, representative good is c , let

$p_{c,t}$: prevailing price of good c in period t

$q_{c,t}$: represents the quantity of c consumed in period t

Let t_n denote current time, and t_0 denote the base year

$$\text{Laspeyres Index} = P_L = \frac{p_{c_1, t_n} q_{c_1, t_0} + p_{c_2, t_n} q_{c_2, t_0} + \dots + p_{c_n, t_n} q_{c_n, t_0}}{p_{c_1, t_0} q_{c_1, t_0} + p_{c_2, t_0} q_{c_2, t_0} + \dots + p_{c_n, t_0} q_{c_n, t_0}} = P_L(t_0, t_n, q_{c, t_0})$$

Fix the basket
at the base year t_0

$$\text{Paasche Index} = P_P = \frac{p_{c_1, t_n} q_{c_1, t_n} + p_{c_2, t_n} q_{c_2, t_n} + \dots + p_{c_n, t_n} q_{c_n, t_n}}{p_{c_1, t_0} q_{c_1, t_n} + p_{c_2, t_0} q_{c_2, t_n} + \dots + p_{c_n, t_0} q_{c_n, t_n}} = P_P(t_0, t_n, q_{c, t_n})$$

Fix the basket at
the current year t_n

In general, a price-index depends:

- Base year
- Current year
- Basket

$$\text{Price Index} = P(t_0, t_n, q_c)$$

$$\frac{P_p \leq P_I \leq P_L}{\downarrow}$$

Potential Remedy is to compute a chain weighted price index
 ↓ can be done with either the Paasche or Laspeyres index.

$$P_L^{\text{Chain-weighted}} = P_L(t_0, t_1, q_{c, t_0}) \times P_L(t_1, t_2, q_{c, t_1}) \times \dots \times P_L(t_{n-1}, t_n, q_{c, t_{n-1}})$$

$$P_p^{\text{Chain-weighted}} = P_p(t_0, t_1, q_{c, t_1}) \times P_p(t_1, t_2, q_{c, t_2}) \times \dots \times P_p(t_{n-1}, t_n, q_{c, t_n})$$

Both chain-weighted indexes ameliorate the problem of undershooting and overshooting the ideal price-index.

Example of Laspeyres Index: Consumer Price Index (CPI)

↳ Fix a { base year basket
base year }

↓
Inflation calculations
across the world.

Example of a Paasche index:
$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

↳ { Fix a base year
Current year basket }

↗ Nominal GDP: current GDP at current prices

Real GDP: current GDP at base year prices